

Cognizant - 100 Technical MCQ Questions

Cognizant Technology Solutions

For BTech Computer Science / IT Students

Object-Oriented Programming

1. Which of these is a valid example of encapsulation in practice?
 - A) Making instance variables private and exposing getters/setters
 - B) Making all variables public
 - C) Using global variables everywhere
 - D) Avoiding classes altogether
2. In Python, which method is automatically called when an object is created?
 - A) `__new__` then `__init__`
 - B) `__del__`
 - C) `__call__`
 - D) `__str__`
3. Which of the following is NOT a pillar of Object-Oriented Programming?
 - A) Encapsulation
 - B) Polymorphism
 - C) Compilation
 - D) Inheritance
4. Which of the following supports multiple inheritance in Java?
 - A) Classes
 - B) Interfaces
 - C) Constructors
 - D) Packages
5. What is the main difference between an abstract class and an interface (pre-Java 8)?
 - A) Interfaces can have constructors
 - B) Abstract classes cannot have any methods
 - C) Interfaces cannot have method bodies while abstract classes can
 - D) There is no difference
6. Which keyword is used in Java to prevent a class from being inherited?
 - A) `static`
 - B) `final`
 - C) `private`
 - D) `abstract`
7. What is operator overloading?
 - A) Giving additional meaning to an operator for user-defined types
 - B) Removing operators from a language
 - C) A compilation error
 - D) A runtime exception

8. What is the output-behavior difference between shallow copy and deep copy of an object?

- A) Shallow copy duplicates only references to nested objects; deep copy duplicates nested objects too
- B) There is no difference
- C) Deep copy is always faster
- D) Shallow copy always throws an exception

9. What is encapsulation?

- A) Hiding implementation details and exposing only functionality
- B) Creating multiple classes
- C) Compiling code faster
- D) Using loops effectively

10. Method overloading is an example of which type of polymorphism?

- A) Runtime polymorphism
- B) Compile-time polymorphism
- C) Dynamic binding
- D) Late binding

11. An abstract class in Java can have:

- A) Only abstract methods
- B) Only concrete methods
- C) Both abstract and concrete methods
- D) No methods at all

12. Which OOP feature allows a subclass to provide a specific implementation of a method already defined in its superclass?

- A) Overloading
- B) Overriding
- C) Encapsulation
- D) Abstraction

13. What does 'this' keyword refer to in Java/C++?

- A) The parent class
- B) The current object instance
- C) A static variable
- D) The compiler

14. Which of these best defines an interface in Java?

- A) A class with only static methods
- B) A contract of methods a class must implement
- C) A private class
- D) A type of exception

DBMS & SQL

- 15. Which property of a transaction ensures that partial updates are never visible to other transactions?**
- A) Atomicity
 - B) Consistency
 - C) Isolation
 - D) Durability
- 16. What is the purpose of database normalization?**
- A) To reduce data redundancy and improve data integrity
 - B) To make queries slower
 - C) To increase storage duplication
 - D) To remove all keys
- 17. What is a deadlock in the context of a database transaction?**
- A) Two or more transactions waiting indefinitely for resources locked by each other
 - B) A syntax error in SQL
 - C) A missing index
 - D) A type of join
- 18. What is a correlated subquery?**
- A) A subquery that references a column from the outer query
 - B) A subquery that never executes
 - C) A subquery used only in INSERT statements
 - D) A subquery without a SELECT clause
- 19. Which SQL command is used to change the structure of an existing table?**
- A) UPDATE
 - B) ALTER
 - C) MODIFY
 - D) CHANGE
- 20. What does ACID stand for in DBMS transactions?**
- A) Atomicity, Consistency, Isolation, Durability
 - B) Accuracy, Consistency, Integration, Data
 - C) Atomic, Complete, Independent, Durable
 - D) Association, Consistency, Isolation, Data
- 21. Which of the following commands removes all rows from a table but keeps its structure, and cannot be rolled back in most databases?**
- A) DELETE
 - B) DROP
 - C) TRUNCATE
 - D) REMOVE
- 22. Which of the following best explains 'eventual consistency' in distributed/NoSQL databases?**
- A) All nodes are always immediately consistent
 - B) Given enough time without new updates, all nodes will converge to the same value
 - C) Data is never consistent
 - D) It refers only to backup consistency

23. In SQL, what does the LIKE operator do?

- A) Performs pattern matching using wildcards such as % and _
- B) Compares two numeric values
- C) Joins two tables
- D) Groups rows

24. Which type of JOIN returns all rows from the left table and matched rows from the right table?

- A) INNER JOIN
- B) LEFT JOIN
- C) RIGHT JOIN
- D) CROSS JOIN

25. Which SQL command grants specific privileges to a user?

- A) REVOKE
- B) GRANT
- C) ALLOW
- D) PERMIT

26. What does 'sharding' mean in database scaling?

- A) Splitting a large database horizontally across multiple servers
- B) Encrypting data
- C) Creating a backup
- D) Normalizing tables

27. Which SQL clause is used to filter groups after a GROUP BY?

- A) WHERE
- B) HAVING
- C) FILTER
- D) ORDER BY

28. Which normal form removes partial dependency on a composite primary key?

- A) 1NF
- B) 2NF
- C) 3NF
- D) BCNF

29. Which index type stores data rows in the same order as the index?

- A) Non-clustered index
- B) Clustered index
- C) Bitmap index
- D) Hash index

30. What is denormalization?

- A) The process of adding redundancy to improve read performance
- B) Removing all keys from a table
- C) A type of join
- D) Encrypting a database

31. A foreign key is used to:

- A) Uniquely identify a row in its own table
- B) Establish a link between two tables
- C) Encrypt sensitive data
- D) Speed up all queries automatically

32. What does DBMS stand for?

- A) Database Management System
- B) Data Backup Management Service
- C) Database Model Structure
- D) Data Base Modeling Software

33. Which of the following is an example of a NoSQL database?

- A) MySQL
- B) PostgreSQL
- C) MongoDB
- D) Oracle

34. What is a view in SQL?

- A) A physical copy of a table
- B) A virtual table based on the result of a SQL query
- C) A type of index
- D) A stored procedure

35. What is a primary key?

- A) A column that can contain duplicate values
- B) A column (or set of columns) that uniquely identifies each row and cannot be NULL
- C) An index used only for sorting
- D) A key used only in foreign tables

36. What is the main advantage of using an ORM (Object-Relational Mapping) tool?

- A) It eliminates the need for a database entirely
- B) It lets developers interact with the database using objects instead of raw SQL
- C) It automatically normalizes data
- D) It replaces indexes

37. What is a materialized view?

- A) A view whose result set is physically stored and periodically refreshed
- B) A temporary table only
- C) A synonym for an index
- D) A type of trigger

38. Which SQL aggregate function returns the number of rows matching a condition?

- A) SUM()
- B) COUNT()
- C) AVG()
- D) MAX()

Operating Systems

39. What is virtual memory?

- A) Physical RAM only
- B) A memory management technique that gives an application the illusion of a large, contiguous memory using disk space
- C) A type of cache memory
- D) A backup memory device

40. What is the difference between a process and a thread?

- A) A process is a lightweight version of a thread
- B) A thread is an independent execution unit within a process, sharing the process's memory space
- C) Threads have separate memory spaces from their process
- D) There is no difference

41. What is the purpose of a semaphore?

- A) To manage access to shared resources and avoid race conditions
- B) To compile code
- C) To manage file names
- D) To render graphics

42. Which page replacement algorithm suffers from Belady's Anomaly?

- A) LRU
- B) FIFO
- C) Optimal
- D) LFU

43. Which of the following is a preemptive scheduling algorithm?

- A) FCFS
- B) Round Robin
- C) Non-preemptive SJF
- D) Batch scheduling

44. What does 'thrashing' refer to in an OS?

- A) Excessive CPU usage by a single process
- B) A state where the system spends most of its time swapping pages instead of executing processes
- C) A type of deadlock
- D) A network congestion issue

45. What is a deadlock?

- A) A state where two or more processes are waiting indefinitely for resources held by each other
- B) A process that has finished execution
- C) A type of memory leak
- D) A CPU scheduling technique

46. Which of the following is NOT one of the four necessary conditions for deadlock?

- A) Mutual Exclusion
- B) Hold and Wait
- C) Preemption
- D) Circular Wait

47. Which of the following synchronization tools is used to prevent race conditions?

- A) Compiler
- B) Semaphore
- C) Scheduler
- D) Loader

48. What is a race condition?

- A) When multiple processes/threads access shared data concurrently and the outcome depends on timing
- B) A process running faster than expected
- C) A type of scheduling algorithm
- D) A network protocol

49. What is the primary function of an operating system?

- A) To compile programs
- B) To manage hardware resources and provide services to applications
- C) To design websites
- D) To encrypt files only

50. What is the difference between a hard link and a symbolic link in a file system?

- A) A hard link points directly to the inode; a symbolic link is a separate file containing a path to the target
- B) They are identical
- C) Symbolic links cannot be broken
- D) Hard links can span across different file systems while symbolic links cannot

51. Which scheduling algorithm can cause starvation?

- A) Round Robin
- B) Priority Scheduling (without aging)
- C) FCFS
- D) None of these

52. Which memory management technique divides memory into fixed-size blocks?

- A) Segmentation
- B) Paging
- C) Swapping
- D) Caching

Computer Networks

53. What is the maximum number of usable host addresses in a /24 subnet?

- A) 256
- B) 254
- C) 255
- D) 512

54. What is the default port number for HTTP?

- A) 21
- B) 25
- C) 80
- D) 443

55. What is the purpose of the ARP protocol?

- A) To resolve an IP address to a MAC address
- B) To resolve a domain name to an IP address
- C) To transfer files
- D) To send email

56. What is the default port number for HTTPS?

- A) 80
- B) 21
- C) 443
- D) 22

57. Which of the following best describes a subnet mask?

- A) It divides an IP address into network and host portions
- B) It encrypts network traffic
- C) It assigns MAC addresses
- D) It is used only in IPv6

58. What does DHCP stand for?

- A) Dynamic Host Configuration Protocol
- B) Data Host Control Protocol
- C) Direct Host Connection Protocol
- D) Domain Host Configuration Protocol

59. Which protocol is used to resolve a domain name to an IP address?

- A) FTP
- B) DNS
- C) SMTP
- D) ARP

60. Which of the following devices operates primarily at the Network layer to forward packets between different networks?

- A) Switch
- B) Hub
- C) Router
- D) Repeater

61. Which protocol operates at the Transport layer and is connection-oriented?

- A) UDP
- B) IP
- C) TCP
- D) ICMP

62. What is the main difference between TCP and UDP?

- A) TCP is connectionless, UDP is connection-oriented
- B) TCP is connection-oriented and reliable, UDP is connectionless and faster but unreliable
- C) There is no difference
- D) UDP guarantees ordered delivery while TCP does not

Data Structures & Algorithms

63. Which sorting algorithm is considered stable and has $O(n \log n)$ time complexity in all cases?

- A) QuickSort
- B) Merge Sort
- C) Selection Sort
- D) Heap Sort

64. Which technique explores all possible options and backtracks when a solution path fails, often used in puzzles like N-Queens?

- A) Greedy algorithm
- B) Backtracking
- C) Dynamic programming
- D) Binary search

65. What is the time complexity of Floyd-Warshall algorithm for all-pairs shortest paths?

- A) $O(V^2)$
- B) $O(V^3)$
- C) $O(E \log V)$
- D) $O(V + E)$

66. What is the amortized time complexity of appending an element to a dynamic array (like Python's list or Java's ArrayList)?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$ amortized
- D) $O(n^2)$

67. What is the space complexity of an iterative approach to computing Fibonacci numbers with memoization storing all n results?

- A) $O(1)$
- B) $O(n)$
- C) $O(n^2)$
- D) $O(\log n)$

68. Which traversal of a Binary Search Tree gives elements in sorted order?

- A) Pre-order
- B) Post-order
- C) In-order
- D) Level-order

69. Which data structure uses FIFO (First In First Out) order?

- A) Stack
- B) Queue
- C) Tree
- D) Graph

70. What is the primary difference between a greedy algorithm and dynamic programming?

- A) Greedy makes a locally optimal choice at each step without reconsidering, DP considers overlapping subproblems and often guarantees a globally optimal solution
- B) They are the same technique
- C) DP never uses recursion
- D) Greedy always gives an optimal solution while DP does not

71. Which data structure uses LIFO (Last In First Out) order?

- A) Queue
- B) Stack
- C) Linked List
- D) Array

72. What is the time complexity of accessing an element in an array by index?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$
- D) $O(n \log n)$

73. What is tail recursion?

- A) A recursive call that is the last operation in a function, which can be optimized by some compilers into iteration
- B) A recursive function with no base case
- C) A function that calls itself twice
- D) A type of infinite loop

74. What is the time complexity of binary search on a sorted array of n elements?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(n \log n)$
- D) $O(1)$

75. Which of the following best describes a trie data structure?

- A) A tree-like structure used for efficient retrieval of strings, especially for prefix matching
- B) A type of balanced binary search tree
- C) A hashing algorithm
- D) A graph traversal method

76. What is the worst-case time complexity of QuickSort?

- A) $O(n \log n)$
- B) $O(n^2)$
- C) $O(\log n)$
- D) $O(n)$

77. Which of the following is true about a doubly linked list compared to a singly linked list?

- A) It uses less memory per node
- B) Each node has pointers to both the next and previous nodes, allowing traversal in both directions
- C) It cannot be traversed backward
- D) It has no advantages

78. What is the time complexity of building a heap from an unsorted array (heapify)?

- A) $O(n \log n)$
- B) $O(n)$
- C) $O(\log n)$
- D) $O(n^2)$

79. What is a balanced binary search tree used to prevent?

- A) Overflow errors
- B) Degeneration into a linked list (worst-case $O(n)$ operations)
- C) Memory leaks
- D) Integer overflow

80. Which of these best describes the 'sliding window' technique?

- A) Maintaining a subset of elements (a window) over a sequence and adjusting its bounds to solve subarray/substring problems efficiently
- B) A sorting technique for linked lists
- C) A hashing technique
- D) A tree traversal method

Programming Language Fundamentals

81. What does 'immutable' mean for an object (e.g., a String in Java)?

- A) Its value can be changed after creation
- B) Its state cannot be changed after it is created
- C) It can only be used once
- D) It cannot be passed to functions

82. What is the purpose of a 'static' variable in a class (Java/C++)?

- A) It belongs to the instance and is unique per object
- B) It belongs to the class rather than any instance and is shared across all instances
- C) It can only be used inside loops
- D) It is always private

83. What is the difference between a shallow copy and a deep copy of a list containing nested lists?

- A) A shallow copy also copies nested objects recursively, deep copy does not
- B) A shallow copy copies references to nested objects, deep copy recursively copies nested objects too
- C) They behave identically
- D) Deep copy is not possible in most languages

84. In Java, what is autoboxing?

- A) Automatic conversion between primitive types and their corresponding wrapper classes
- B) A way to box multiple objects together
- C) A garbage collection technique
- D) A type of casting error

85. Which of the following is a value type (not a reference type) in Java?

- A) int
- B) String
- C) Array
- D) Object

86. In C, what does the 'malloc' function do?

- A) Deallocates memory
- B) Allocates a block of memory dynamically
- C) Initializes a variable to zero
- D) Declares a static variable

87. In C++, what does the 'const' keyword do when applied to a variable?

- A) Makes the variable a pointer
- B) Prevents the variable's value from being modified after initialization
- C) Declares a static variable
- D) Declares a global variable

88. What is duck typing (commonly associated with Python)?

- A) A type system that checks types strictly at compile time
- B) An object's suitability is determined by the presence of certain methods/behavior rather than its explicit type
- C) A type of exception
- D) A memory allocation strategy

89. What is method chaining?

- A) Calling multiple methods on the same object in a single statement by having each method return the object itself
- B) A way to declare multiple classes
- C) A type of exception handling
- D) A memory management technique

90. What is the difference between a compiler and an interpreter?

- A) A compiler translates the whole program to machine code before execution, an interpreter executes code line-by-line
- B) They are identical
- C) Interpreters are always faster than compilers
- D) Compilers cannot detect syntax errors

91. Which of the following is true about Python's Global Interpreter Lock (GIL)?

- A) It allows true parallel execution of threads on multiple cores
- B) It ensures only one thread executes Python bytecode at a time in CPython, limiting CPU-bound multithreading performance
- C) It has no impact on multithreading
- D) It is used only in Python 2

92. What is the default value of an uninitialized boolean instance variable in Java?

- A) true
- B) false
- C) null
- D) 0

Web & Cloud Fundamentals

93. What does CSS stand for?

- A) Cascading Style Sheets
- B) Computer Style Sheets
- C) Creative Style System
- D) Cascading Software Sheets

94. Which HTTP method is typically used to retrieve data without side effects?

- A) POST
- B) GET
- C) DELETE
- D) PUT

95. What is the purpose of cookies in web applications?

- A) To store small pieces of data on the client side to maintain state/session information
- B) To encrypt HTTPS traffic
- C) To speed up DNS resolution
- D) To compress images

96. What is the difference between IaaS, PaaS, and SaaS?

- A) They are all the same cloud service model
- B) IaaS provides raw infrastructure, PaaS provides a platform to build apps, SaaS delivers ready-to-use software
- C) SaaS provides infrastructure while IaaS provides software
- D) PaaS is only for storage

97. What is a RESTful API?

- A) An API that strictly requires SOAP protocol
- B) An architectural style for designing networked applications using stateless operations over standard HTTP methods
- C) A type of database
- D) A programming language

98. What is auto-scaling in cloud computing?

- A) Automatically adjusting compute resources up or down based on demand
- B) Manually adding servers only
- C) A type of database indexing
- D) A billing model only

99. What is the primary difference between HTTP and HTTPS?

- A) HTTPS uses encryption (SSL/TLS) to secure data in transit while HTTP does not
- B) HTTP is faster in all cases
- C) HTTPS cannot use port 443
- D) There is no difference

100. What is the purpose of a Content Delivery Network (CDN)?

- A) To cache and serve content from servers geographically closer to users, reducing latency
- B) To encrypt all web traffic
- C) To replace DNS servers
- D) To manage database transactions

Answer Key

1-A	2-A	3-C	4-B	5-C	6-B	7-A	8-A	9-A	10-B
11-C	12-B	13-B	14-B	15-C	16-A	17-A	18-A	19-B	20-A
21-C	22-B	23-A	24-B	25-B	26-A	27-B	28-B	29-B	30-A
31-B	32-A	33-C	34-B	35-B	36-B	37-A	38-B	39-B	40-B
41-A	42-B	43-B	44-B	45-A	46-C	47-B	48-A	49-B	50-A
51-B	52-B	53-B	54-C	55-A	56-C	57-A	58-A	59-B	60-C
61-C	62-B	63-B	64-B	65-B	66-C	67-B	68-C	69-B	70-A
71-B	72-C	73-A	74-B	75-A	76-B	77-B	78-B	79-B	80-A
81-B	82-B	83-B	84-A	85-A	86-B	87-B	88-B	89-A	90-A
91-B	92-B	93-A	94-B	95-A	96-B	97-B	98-A	99-A	100-A

Format: [Question No.]-[Correct Option Letter]. Content compiled from publicly available 2026 placement-prep resources and established CS-fundamentals question banks - verify current-year specifics before your interview.